TEXORA · LECTURE SHEET

Boiler

The boiler is the heart of the textile utility system: dyeing, finishing, humidification and heating all draw its steam. This sheet covers boiler types, the mountings that make a boiler safe and controllable, the accessories that make it efficient, and how efficiency is measured by the direct and indirect methods.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 3 · LEVEL 4 TERM 1

1. What a boiler is

A **boiler** is a closed pressure vessel in which water is heated by the combustion of fuel (or by recovered waste heat) to produce steam at a required pressure and temperature. Performance is stated as **evaporation capacity** (kg steam/h), **working pressure** (bar), steam condition (dry saturated or superheated) and **efficiency**. Mill boilers in wet processing are commonly oil- or gas-fired package boilers supplying saturated steam to dyeing machines, dryers and calendars, with a standby unit for continuity.

2. Boiler types

2.1 By the position of hot gases and water

- **Fire-tube boiler:** hot gases pass *inside* tubes surrounded by water (e.g. Lancashire, Cochran, package fire-tube). Simple, robust, large water storage — good for fluctuating mill loads — but pressure is limited and a tube failure releases a large stored energy.
- **Water-tube boiler:** water circulates *inside* tubes heated from outside by hot gases (e.g. Babcock & Wilcox type, modern bent-tube designs). Higher pressures and capacities, faster steam raising, safer in failure (small water volume per tube), but needs better water treatment and more skilled care.

2.2 Other classifications

BASIS	TYPES	NOTE
Axis of shell	Horizontal, vertical	Vertical (Cochran-type) saves floor space, small capacity
Furnace position	Internally fired, externally fired	External furnace suits solid fuel with a grate

Circulation	Natural (thermal), forced (pumped)	Forced circulation for high pressure / special designs
Duty / form	Stationary, portable, package	Package boilers arrive assembled — common in mills
Heat source	Fired, waste-heat (HRSG, economiser boilers)	Waste-heat units recover exhaust energy (Topic 4)

3. Boiler mountings

Mountings are fittings mounted directly on the boiler that are essential for **safety and control**; a boiler may not legally be steamed without them.

- **Safety valves (usually two):** lift automatically to release steam when pressure exceeds the set value.
- **Water-level gauge glass:** shows the true water level; two gauges are fitted on larger boilers.
- **Pressure gauge:** shows steam pressure, with a siphon to protect it from steam temperature.
- **Main stop valve:** isolates the boiler from the steam range.
- **Feed check valve:** allows feedwater in but prevents return flow from the boiler.
- **Blow-off (blowdown) valve:** discharges water, sludge and dissolved solids to control concentration.
- **Fusible plug (fire-tube):** melts if the crown is uncovered, warning of low water.
- **Low-water alarm / cut-off** on automatic boilers: stops the burner before damage occurs.

4. Boiler accessories

Accessories are devices added to improve **efficiency and operation**; the boiler can run without them, but wastefully.

- **Economiser:** heats feedwater with flue gas leaving the boiler — every degree of feedwater preheat is fuel saved.
- **Air preheater:** heats combustion air with flue gas, improving combustion and recovering heat.
- **Superheater:** dries and superheats steam for turbines (less used for process steam).
- **Feed pump / injector:** delivers water against boiler pressure.
- **Steam separator, traps and strainers:** deliver dry steam and return condensate.
- **Chimney / induced-draught fan:** creates draught to move gases through the boiler.

	MOUNTINGS	ACCESSORIES
Purpose	Safety and control — mandatory	Efficiency and convenience — optional

Examples	Safety valve, gauge glass, pressure gauge, stop valve, feed check valve, blow-off valve	Economiser, air preheater, superheater, feed pump, steam trap
If missing	Boiler must not be operated	Boiler runs, but fuel cost rises

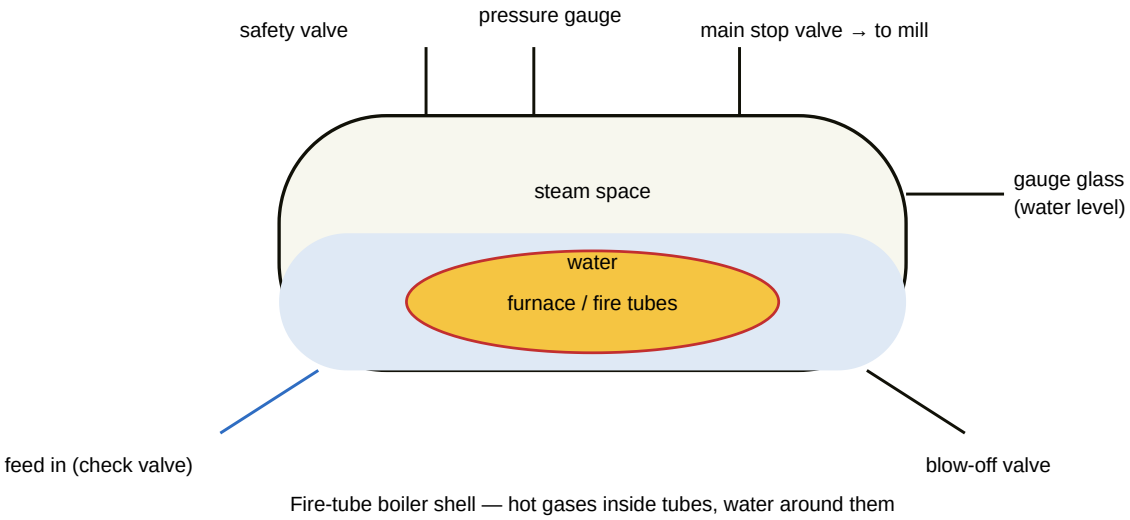


Figure 1 — Fire-tube boiler schematic with the principal mountings labelled: safety valve, pressure gauge, main stop valve, gauge glass, blow-off valve and feed check valve.

5. Boiler efficiency — direct method

Definition. Boiler efficiency is the fraction of fuel heat that appears as useful heat in the steam:

$$\eta\ (\text{direct}) = [\ m_s \times (h_s - h_{fw})\ ] \div [\ m_f \times CV\ ] \times 100\%$$

where m_s = steam raised (kg/h), h_s = steam enthalpy (kJ/kg), h_{fw} = feedwater enthalpy (kJ/kg), m_f = fuel burnt (kg/h), CV = fuel calorific value (kJ/kg).

WORKED EXAMPLE — DIRECT-METHOD EFFICIENCY

Steam raised m_s = 4,800 kg/h; steam enthalpy (dry saturated, 10 bar) h_s = 2,778 kJ/kg; feedwater at 40 °C, h_{fw} = 167.5 kJ/kg; fuel m_f = 600 kg/h; CV = 25,000 kJ/kg.

Useful heat = 4800 × (2778 − 167.5) = 4800 × 2610.5 = 12,530,400 kJ/h.

Fuel heat = 600 × 25,000 = 15,000,000 kJ/h.

η = 12,530,400 / 15,000,000 = 0.835 ≈ **83.5%**.

So about 16.5% of the fuel heat is lost — mainly up the stack. That loss figure is what the indirect method breaks down item by item.

6. Boiler efficiency — indirect method

The **indirect (heat-loss) method** measures the losses and subtracts them from 100%:

$$\eta \text{ (indirect)} = 100\% - (\text{dry flue-gas loss} + \text{moisture/H}_2 \text{ loss} + \text{unburnt fuel loss} + \text{blowdown loss} + \text{radiation loss} + \ldots)$$

Flue-gas temperature and composition (O_2/CO_2, CO) are measured, each loss is calculated from standard formulae, and the largest loss points directly to the corrective action: high stack temperature → fouled surfaces or excess air; CO present → incomplete combustion and wasted fuel; high blowdown → over-concentrated boiler water control failure. The indirect method needs no accurate steam-flow measurement, which is why it is preferred for acceptance tests, while the direct method is preferred for routine day-to-day monitoring when steam and fuel metering exist.

Raising efficiency in practice: recover condensate (hot, treated water returned is fuel saved twice), fit and maintain the economiser, control excess air, insulate steam lines, repair traps and leaks, blow down by measurement (TDS control) rather than by habit, and keep heat-transfer surfaces clean.

Glossary

Fire-tube boiler — hot gases inside tubes, water outside. **Water-tube boiler** — water inside tubes, hot gases outside. **Mountings** — mandatory safety and control fittings on the boiler. **Accessories** — optional devices that raise efficiency. **Economiser** — flue-gas feedwater heater. **Blowdown** — discharge of boiler water to control dissolved solids. **Calorific value (CV)** — heat released per unit mass (or volume) of fuel. **Direct method** — efficiency from measured useful heat ÷ fuel heat. **Indirect method** — 100% minus the sum of measured losses.

Key points

• Fire-tube: simple, large storage, limited pressure. Water-tube: high pressure and capacity, needs pure feedwater.
• Mountings = safety (must have); accessories = efficiency (should have).
• Direct method example: 4800 × (2778 − 167.5) ÷ (600 × 25,000) ≈ 83.5%.
• Indirect method: η = 100% − total losses; it diagnoses *where* the heat went.
• Condensate return, economiser, excess-air control and TDS-based blowdown are the mill's main fuel savers.

Practice

1. A boiler raises 3,000 kg/h of steam with h_s = 2,760 kJ/kg from feedwater of 150 kJ/kg, burning 320 kg/h of fuel of CV 26,000 kJ/kg. Find the direct-method efficiency.
2. An indirect test finds losses: flue gas 9%, moisture 4%, unburnt 1%, blowdown 1%, radiation 2%. Find the efficiency and the largest single improvement target.
3. Classify each as mounting or accessory and give its function: safety valve, economiser, gauge glass, superheater, blow-off valve, feed pump.
4. Why does returning hot condensate raise boiler efficiency in two distinct ways?

Answer key

1. Useful = 3000 × (2760 − 150) = 7,830,000 kJ/h. Fuel = 320 × 26,000 = 8,320,000 kJ/h. η = 7,830,000/8,320,000 ≈ **94.1%** — note this is an idealised exercise value; check metering if a real boiler reports this high.
2. Total loss = 9 + 4 + 1 + 1 + 2 = 17%, so η = **83%**. Largest loss is flue gas (9%) — check excess air and surface fouling first.
3. Safety valve — mounting, releases overpressure. Economiser — accessory, preheats feedwater with flue gas. Gauge glass — mounting, shows water level. Superheater — accessory, superheats steam. Blow-off valve — mounting, discharges sludge/TDS. Feed pump — accessory, delivers feedwater.
4. (1) Its sensible heat re-enters the boiler, so less fuel heat is needed to raise steam. (2) It is already treated, soft water, so make-up treatment cost, blowdown loss and scale risk all fall.

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